



NTNU – Trondheim
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TET4185: Power Markets, Resources and Environment

**Microeconomics-Solutions to the
exercises during the lecture**

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Spring 2025

Solution-Demand Price Elasticity

- Prices, linear model

- $p = 200 - 0.014D$
- $p_{9000} = 74, p_{10000} = 60, p_{11000} = 46$

- Prices, exponential model

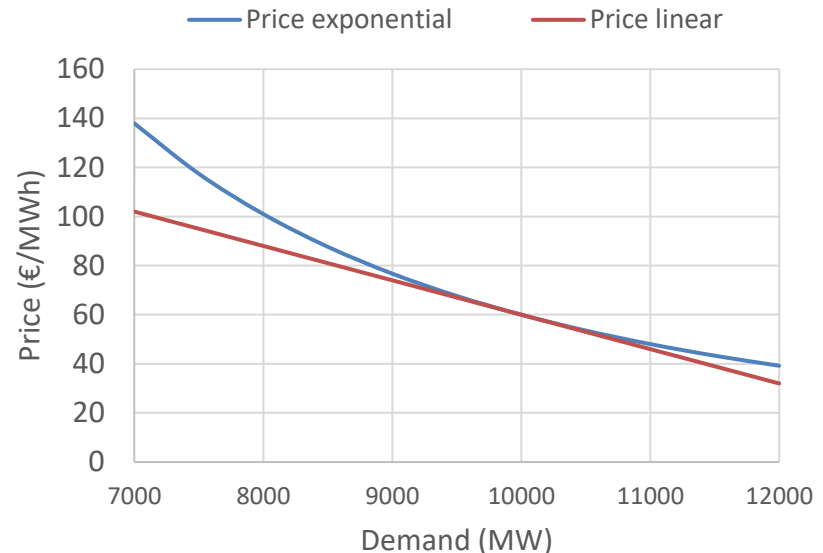
- $p = A^{-1/k} \cdot D^{1/k}$
- $p_{9000} = 76.7, p_{10000} = 60, p_{11000} = 48.0$

- Elasticities, linear model

- $e = \frac{\partial D}{\partial p} \cdot \frac{p}{D} = 1 - \frac{200}{0.014 \cdot D}$
- $e_{9000} = -0.5873, p_{10000} = -0.4286, p_{11000} = 0.2987$

- Elasticities, exponential model

- $e = -0.4286$ (constant)



Solution-Competition

- Competitive solution: easy to see from figure, $P = 40 \text{ €/MWh}$

$$D = \frac{200 - 40}{0.014} = 11428.5 \text{ MW}$$

- $\pi = 11428.5 \cdot 40 - 1428.5 \cdot 40 = 400.000 \text{ €}$

- Alternative solution (revenue):

- Revenue: $\pi = P \cdot X - C(X)$

- With $D = X$:
$$\frac{d\pi}{dX} = 0 \Rightarrow 200 - 0.014X = c$$

where

$$c = 0 \text{ for } 0 \leq X \leq 10000$$

$$c = 40 \text{ for } 10000 \leq X \leq 13000$$

$$c = 45 \text{ for } 13000 \leq X \leq 16000$$

Solution-Competition

- Monopoly solution: assume one producer owns everything

$$\pi = P(X) \cdot X - C(X)$$

With $D = X$:

$$\frac{d\pi}{dX} = 0 \Rightarrow 200 - 0.028X = c$$

where

$$c = 0 \text{ for } 0 \leq X \leq 10000$$

$$c = 40 \text{ for } 10000 \leq X \leq 13000$$

$$c = 45 \text{ for } 13000 \leq X \leq 16000$$

- results in $Q = D = 7142.8$ MW and $P = 100$ €/MWh
- $\pi = 7142.8 \cdot 100 = 714280$ €

Solution-Competition

- Cournot solution: assume hydro produces anyway, and two others compete

$$\pi_i = P \cdot q_i - c_i q_i$$

$$\frac{\partial \pi_i}{\partial q_i} = P + q_i \frac{\partial P}{\partial q_i} - c_i = 0$$

this gives:

$$P - 0.014q_2 - 40 = 0$$

$$P - 0.014q_3 - 45 = 0$$

results in:

$$200 - 0.014 \cdot (10000 + q_2 + q_3) - 0.014q_2 - 40 = 0$$

$$200 - 0.014 \cdot (10000 + q_2 + q_3) - 0.014q_3 - 45 = 0$$

this becomes:

$$2q_2 + q_3 = 1428.6$$

$$q_2 + 2q_3 = 1071.4$$

and

$$q_2 = 238.1, q_3 = 595.2, Q = 10833.3 \text{ MW and } P = 48.3 \text{ €/MWh}$$